Alternative Model Systems for the Major Depressive Disorder Pathograph

Disease: Major Depressive Disorder (MONDO:0002009) **Brief:** Identify model systems — beyond the six already curated — that could legitimately support specific, named pathograph nodes, and classify each by whether it recapitulates, perturbs, is informative by analogy only, or fails to recapitulate the node it is mapped to.

Summary

Beyond the six curated *C. elegans* and mouse models, a large field of uncurated systems can legitimately support MDD pathograph nodes — but their value is sharply **stratified by node**, and the most interesting focus nodes are precisely the ones that resist substitution. The strongest, cleanest additions are for the **Oligodendrocyte Dysfunction and Demyelination** node (cuprizone and lysolecithin mice, genetically tractable larval and `mbp:NTR` zebrafish, and human iPSC myelinating co-cultures) and for the **Neuroinflammation** node (human iPSC microglia, microglia-containing cerebral organoids, BBB-on-chip and neurovascular-unit platforms, adult chronic-stress zebrafish, and human IFN-α therapy / LPS endotoxin challenge). For **HPA Axis Dysregulation** the zebrafish glucocorticoid-receptor mutant and GRE reporter line, the tree shrew, and the cynomolgus monkey are all defensible, uncurated candidates. For **Stress-Responsive Transcription Factor Reprogramming**, the glucocorticoid-responsiveness of KLF9, KLF15 and EGR1 is conserved all the way to *Xenopus* and reproducible in rodent and in-vitro reporter systems, so these systems can *perturb* the transcriptional step the node names.

The three focus nodes that were defined from **human post-mortem hippocampus** — PENK+ GABAergic Interneuron Activation, Neurogenic Niche Interferon Signaling Activation, and the transcription- factor reprogramming module — behave very differently under substitution. Here the question is not whether another taxon shares a mammalian structure, but whether any model can host a mechanism first *observed in human tissue*. For these, most candidate systems are honestly classified as **(b) perturbs** (they reproduce a molecular or cellular step) or **(c) informative by analogy only** (the defining feature is absent). In particular, **the PENK+ GABAergic hippocampal interneuron node has no model that recapitulates it**: an

enkephalinergic GABAergic cell type exists in mammalian neocortex and as striatal medium spiny neurons, but the specific hippocampal population the node names — described as not previously reported in the human hippocampus — has not been identified in any model system.

The honest bottom line: **a short report that correctly rules candidates out is more useful than a long one that overstates them.** Drosophila and additional *C. elegans* models are empty for all five focus nodes (they do not myelinate and lack the relevant cell types); teleost adult neurogenesis is the opposite of the "stalled human" phenotype and so is analogy-only for the neurogenesis node; and the PENK+ hippocampal interneuron remains genuinely unsubstitutable. The strongest non-mouse additions cluster where the biology is evolutionarily conserved (myelin, type I interferon, the glucocorticoid axis, KLF/EGR transcription factors) and the human-defined single-nucleus modules remain only partially addressable.

Key Findings

Finding 1 — HPA Axis Dysregulation: the zebrafish glucocorticoid-receptor mutant is a strong, uncurated AnimalModel

The glucocorticoid-receptor (`gr`) loss-of-function zebrafish (*Danio rerio*) generated by Ziv et al. is an adult-viable mutant in which negative feedback on the stress response is abolished: cortisol is elevated but cannot signal through GR, producing freezing, reduced exploration and failure to habituate — behaviours reversed by fluoxetine. Teleosts possess a conserved hypothalamic–pituitary–interrenal (HPI) cortisol axis and a GR orthologue, so this model **(a) recapitulates** the HPA Axis Dysregulation node at the level of disrupted glucocorticoid feedback. A companion transgenic GRE-reporter line (SR4G) provides single-cell GR-activity readout throughout the brain, enabling mechanistic perturbation rather than observation alone.

This would be curated as an **AnimalModel**. Its limitation is important and must be stated plainly: fish use cortisol via an interrenal (HPI) axis rather than an adrenal HPA axis, lack a laminated mammalian hippocampus, and the readout is behavioural/endocrine — **not** the human single-nucleus RNA-seq transcriptional modules that define the hippocampus-derived focus nodes. The model carries the stress-axis lesion but not the human hippocampal molecular signature.

> *"Dysregulation of the stress axis is a hallmark of major depression in human patients."* and *"We identified an adult-viable zebrafish mutant in which the negative feedback on the stress response is disrupted, due to abolition of all transcriptional activity of GR." — An affective disorder in zebrafish with mutation of the glucocorticoid receptor*, PMID: 22641177

Finding 2 — Stress-Responsive Transcription Factor Reprogramming: KLF9/KLF15/EGR1 are conserved, directly perturbable glucocorticoid targets

Node 11 (Stress-Responsive Transcription Factor Reprogramming, which I take to be a **MOLECULAR** node per its description) names KLF9, KLF15, EGR1 and relatives. Multiple uncurated systems can **(b) perturb** this node by reproducing the glucocorticoid → KLF/EGR transcriptional step:

- **Mouse hippocampus and the HT-22 hippocampal cell line** (Bagamasbad et al.): corticosterone directly induces *Klf9* via two GR/MR response elements, and the −5.3 kb GRE/MRE is largely conserved across tetrapods. The induction is protein-synthesis-independent, establishing KLF9 as a *direct* glucocorticoid-response gene. HT-22 would be an **ExperimentalModel**; the mouse an **AnimalModel**.
- **Rat hippocampus** (Kennedy et al.): genome-wide ChIP-seq maps KLF-family members as MR/GR targets after acute stress — an **AnimalModel**.
- **Xenopus tropicalis** GR-knockout tadpoles: corticosterone induction of *klf9* is completely abrogated, and the *Xenopus* XLK-WG cell line shows CORT-induced *klf9* via a synergy module whose AHRE cluster is positionally conserved in the human genome — **AnimalModel** (whole tadpole) and **ExperimentalModel** (XLK-WG line).
- **EGR1 systems**: in neuronal BZ cells and primary neurons, GR physically interacts with an EGR1 response element to repress BDNF exon IV (linking node 11 to **Neuroplasticity Deficits**); in PC12 cells dexamethasone blunts IL-6-induced *Egr1*/*c-Fos*. Both **ExperimentalModels**.

The shared limitation: none reconstitutes the human hippocampal multi-cell module or the PENK+ interneuron context in which the module was defined, and the *Xenopus* readout is metamorphosis, not mood.

> *"The CORT induction of Klf9 was resistant to protein synthesis inhibition, suggesting that Klf9 is a direct CORT-response gene."* and *"The -5.3-kb GRE/MRE is largely conserved across tetrapods" — Molecular basis for glucocorticoid induction of the Kruppel-like factor 9 gene in hippocampal neurons*, PMID: 22962255
>
> *"Induction by exogenous corticosterone (CORT, the frog stress hormone) of the CORT response genes, klf9 ... was completely abrogated in GRKO tadpoles." — Glucocorticoid receptor is required for survival through metamorphosis in the frog Xenopus tropicalis*, PMID: 32032606
>
> *"both GR and EGR1 bind to this promoter sequence in a glucocorticoid-dependent manner" — Crosstalk Between Glucocorticoid Receptor and Early-growth Response Protein 1 Accounts for Repression of Brain-derived Neurotrophic Factor Transcript 4 Expression*, PMID: 30578973

Finding 3 — Oligodendrocyte Dysfunction and Demyelination: three complementary uncurated models, with invertebrates ruled out

This node (which I take to be **CELLULAR/TISSUE** level per its description) has the richest set of defensible, uncurated additions:

1. **Cuprizone mouse** (**AnimalModel**): the copper chelator causes corpus-callosum demyelination accompanied by anxiety- and depression-like behaviour. Classification **(a) recapitulates** the demyelination lesion — but the readout couples a behavioural proxy to myelin loss, which must not be conflated with the human transcriptional signature.
2. **Larval zebrafish** (**AnimalModel**): genetically tractable, live-imageable CNS myelination; the `Tg(mbp:mCherry-NTR)` line enables inducible, reversible demyelination/remyelination. Classification **(a) recapitulates** oligodendrocyte biology at single-cell resolution.
3. **Human iPSC neuron–oligodendrocyte co-culture** (**ExperimentalModel**): human O4+ oligodendrocytes myelinate axons in vitro. Classification **(b) perturbs / recapitulates** human myelination but lacks the in-vivo stress/HPA context.

Critically, **invertebrates (Drosophila, C. elegans) are ruled out for this node** — they do not myelinate, so no manipulation in them can host the mechanism. This is a clean negative and a useful result. A lysolecithin/LPC focal demyelination mouse is a further mammalian **AnimalModel** not curated on this entry.

> *"As a model organism, larval zebrafish have been extensively employed to investigate the molecular and cellular basis of CNS myelination, because of their genetic tractability and suitability for non-invasive live cell imaging." — CNS Hypomyelination Disrupts Axonal Conduction and Behavior in Larval Zebrafish*, PMID: 34544838
>
> *"Neurons form synaptic connections and transmit signals, while oligodendrocytes, including oligodendrocyte precursor cells (OPCs) and their derivatives, are vital for central nervous system maintenance and myelination." — Human iPSC-Derived Neuron and Oligodendrocyte Co-culture as a Small-Molecule Screening Assay for Myelination*, PMID: 40364981

Finding 4 — Neurogenic Niche Interferon Signaling + Stalled Adult Hippocampal Neurogenesis: mechanistically hostable, but the niche is human-hippocampus-defined

Node 10 (Neurogenic Niche Interferon Signaling Activation, **MOLECULAR**) and node 9 (Stalled Adult Hippocampal Neurogenesis, **CELLULAR**) have no curated model. The candidates split cleanly:

- **Mouse adult neural stem cells**: primed quiescent NSCs carry an intrinsic interferon-response signature, and aging niche-derived inflammatory signals impose quiescence. These show the type I interferon pathway operating in the mammalian neurogenic niche to restrain neurogenesis — classification **(b) perturbs** node 10 and **(a) recapitulates** the stalled-neurogenesis phenotype — but in the SVZ rather than the human hippocampal niche defined by snRNA-seq. Curated as **AnimalModel** / primary culture.
- **Teleost fish (zebrafish, killifish)**: constitutive adult neurogenesis across many niches — classification **(c) informative by analogy only** for node 9. Fish have no hippocampus (they use pallium) and robust lifelong neurogenesis, the *opposite* of the contested, low-rate human hippocampal phenotype, so they cannot reproduce the "stalled human" condition. However, type I interferon is vertebrate-conserved in teleosts (confirmed in zebrafish via FTR33 regulation of type I IFN/ISGs), so fish can host node 10 *mechanistically* but not in a hippocampal context.

> *"Primed quiescent NSCs (p-qNSCs) are defined by high levels of LRIG1 and CD9, as well as an interferon response signature" — LRIG1 is a gatekeeper to exit from quiescence in adult neural stem cells*, PMID: 33972529

> *"zebrafish exhibit constitutive neurogenesis across multiple stem cell niches that provide new neurons to every major brain division" — The role of neuro-epithelial-like and radial-glial stem and progenitor cells in development, plasticity, and repair*, PMID: 29902500

Finding 5 — PENK+ GABAergic Interneuron Activation: no model recapitulates; homologous cell types exist but not the hippocampal population

Node 12 (**CELLULAR**) describes a PENK+ GABAergic interneuron population explicitly "not previously reported in the human hippocampus." Searching for proenkephalin/enkephalin GABAergic hippocampal populations, I found that enkephalinergic GABAergic interneurons are well documented in rodent **neocortex** (a VIP+/enkephalin+ mu-opioid-sensitive interneuron class; preproenkephalin- immunoreactive GABAergic cortical interneurons) and as **striatal medium spiny neurons** (a GABAergic projection-neuron population expressing proenkephalin). These establish that a homologous PENK+/GABAergic cell *type* exists in mammals — but in neocortex and striatum, not the hippocampal interneuron population the node names.

Human iPSC-derived MGE/LHX6+ GABAergic interneurons (**ExperimentalModel**) can generate diverse neuropeptide+ interneuron subtypes but no study has demonstrated the specific hippocampal PENK+ population. **Classification (c) informative by analogy only**: the enkephalinergic GABAergic module exists and can be perturbed (opioid/enkephalin manipulation), but the node's defining hippocampal population has not been identified in any model system. This is the single node where no candidate reaches **(a)**, and that is the honest answer.

> *"mu-opioid receptor-immunoreactive neurons very frequently exhibited preproenkephalin immunoreactivity" — A group of cortical interneurons expressing mu-opioid receptor-like immunoreactivity: a double immunofluorescence study in the rat cerebral cortex*, PMID: 10854753
>
> *"layers II-V cells expressing vasoactive intestinal peptide and enkephalin (Enk)" — Extensive overlap of mu-opioid and nicotinic sensitivity in cortical interneurons*, PMID: 17068095
>
> *"In the neostriatum, the proenkephalin gene is expressed in medium spiny GABA neurons, which project to the globus pallidus." — Intrinsic GABA neurons inhibit proenkephalin gene expression in slice cultures of rat neostriatum*, PMID: 11982623

Finding 6 — Neuroinflammation: human iPSC microglia, microglia-in-organoids, BBB-chips, and human IFN-α/LPS challenge add non-mouse models

The Neuroinflammation node is curated only with mouse models. Uncurated additions span three schema situations:

- **Human iPSC-derived microglia-like cells** (**ExperimentalModel**): mount stimulus-specific inflammatory responses to TNF-α, IFN-γ, LPS and Poly(I:C) — **(a) recapitulates** human microglial activation. Type-I-interferon signalling drives microglial dysfunction/senescence in iPSC models, directly bridging to node 10.
- **Microglia-containing cerebral organoids** (**ExperimentalModel**): respond to inflammatory stimuli including IFN-γ — **(a/b)**, the relevant immune cell type is present and arises.
- **Human BBB-on-chip / neurovascular-unit chips** (**ExperimentalModel**): model cytokine (TNF-α/LPS)-driven barrier disruption and immune-cell migration — **(b) perturbs** the cytokine-entry/BBB step.
- **Human IFN-α therapy** (for hepatitis C) and **low-dose LPS endotoxin challenge** in volunteers: these are **human experimental/iatrogenic models**, not animal or in-vitro models. They would *not* fit the AnimalModel block, and as controlled-exposure/observational human data they do not map onto either the AnimalModel or ExperimentalModel block cleanly — they are human induced models of inflammation-associated depression.
- **Adult zebrafish chronic unpredictable stress** (**AnimalModel**): couples depression-like behaviour to microglial/astroglial neuroinflammation markers, cortisol and serotonin turnover — a non-mouse addition for this node and for HPA Axis Dysregulation.

> *"MGLCs displayed distinct morphologies and produced stimulus- and time-dependent cytokine secretion profiles upon exposure to diverse inflammatory stimuli, including pro-inflammatory cytokines (TNFα, interferon-γ) and agonists of the Toll-like receptors" — Differential Responses of Human iPSC-Derived Microglia to Stimulation with Diverse Inflammogens*, PMID: 41227333
>
> *"Cytokine treatment with Interferon-alpha (IFN-α) represents a clinical model of immune associated depression" — Influence of gender on cytokine induced depression and treatment*, PMID: 34171555

Finding 7 — Patient-derived and defined-genetic human in-vitro systems close the non-focus-node gaps (E-I imbalance, mitochondrial, neuroplasticity)

Several uncurated human in-vitro systems address nodes that had no models:

- **Patient-derived SSRI-resistant MDD iPSC serotonergic neurons** (**ExperimentalModel**): show altered neurite growth/morphology downstream of lowered Protocadherin alpha genes — maps to **Neuroplasticity Deficits** (and Monoamine Deficiency); **(a)** for a human patient-derived MDD readout.
- **DISC1-mutant iPSC glutamatergic/GABAergic co-cultures** (**ExperimentalModel**): show aberrant GABAergic synapse densities — maps to **Excitation-Inhibition Imbalance** (node 2, scale **not stated**; I take it to be CELLULAR/circuit level); **(b)**.
- **TSPO-knockout hiPSC neurons/NPCs/astrocytes** (**ExperimentalModel**): reduced mitochondrial respiration and glycolysis — maps to **Mitochondrial Dysfunction** (node 6); **(b)**.
- **Perturbable adult human organotypic slice cultures** and **fetal human organotypic cortical cultures with shRNA knockdown** (**ExperimentalModel**): fill the previously empty human post-mortem/surgical-tissue category and allow genetic perturbation of human tissue affecting E-I / circuit function.

> *"NR patient-derived serotonergic neurons exhibited altered neurite growth and morphology downstream of lowered expression of key Protocadherin alpha genes as compared to healthy controls and Rs." — Altered serotonergic circuitry in SSRI-resistant major depressive disorder patient-derived neurons*, PMID: 30903001
>
> *"Human organotypic slice cultures provide experimental access to adult human neuronal circuits ex vivo" — Adaptive Network Stabilization Despite Profound Structural Reorganization in Human Organotypic Cultures*, PMID: 42750245
>
> *"STXBP1 knockdown disrupted spontaneous subplate neuronal network activity observed by calcium imaging at 14 days in vitro, reducing signal amplitude and synchronicity, indicating impaired early circuit function." — Human brain slice culture model of STXBP1 encephalopathy reveals disruption of neuronal network activity and changes in gene expression*, PMID: 42214744

Candidate Model Inventory (consolidated)

The table below consolidates every proposed candidate with its node mapping, classification, schema block, primary PMID, and the key limitation that constrains it.

System (species / platform)	Category	Node(s) mapped	Class	Schema block	PMID
Danio rerio `gr` mutant + SR4G GRE reporter	Genetic	HPA Axis Dysregulation	(a)	AnimalModel	22641177, 24679220
Mouse HT-22 line / mouse hippocampus (CORT→Klf9)	Perturbation	Stress-Responsive TF Reprogramming	(b)	ExperimentalModel / AnimalModel	22962255
Rat hippocampus MR/GR ChIP-seq (KLFs)	Perturbation	Stress-Responsive TF Reprogramming	(b)	AnimalModel	36942087
Xenopus tropicalis GR-KO tadpole; XLK-WG line	Genetic / Perturbation	Stress-Responsive TF Reprogramming	(b)	AnimalModel / ExperimentalModel	32032606, 35172007
BZ cells / PC12 (GR–EGR1, Egr1/c-Fos)	Perturbation	Stress-Responsive TF Reprogramming; Neuroplasticity	(b)	ExperimentalModel	30578973, 33454999
Cuprizone mouse	Controlled exposure	Oligodendrocyte Dysfunction & Demyelination	(a)	AnimalModel	38394854
Larval zebrafish; `mbp:NTR` ablation line	Genetic	Oligodendrocyte Dysfunction & Demyelination	(a)	AnimalModel	34544838, 30820902
Human iPSC neuron–oligodendrocyte co-culture	Engineered in vitro	Oligodendrocyte Dysfunction & Demyelination	(b)	ExperimentalModel	40364981
Mouse adult NSCs (IFN-	Observational / Perturbation	Neurogenic Niche IFN Signaling;	(b)/ (a)	AnimalModel / primary culture	33972529, 30827680

System (species / platform)	Category	Node(s) mapped	Class	Schema block	PMID
response, quiescence)		Stalled Neurogenesis			
Teleost (zebrafish/ killifish) adult neurogenesis	Observational	Stalled Adult Hippocampal Neurogenesis	(c)	AnimalModel	29902500
Zebrafish type I IFN (FTR33/ ISGs)	Genetic	Neurogenic Niche IFN Signaling	(b)	AnimalModel	36801427
Rodent neocortical enkephalin+ GABAergic interneurons	Observational	PENK+ GABAergic Interneuron Activation	(c)	AnimalModel / ex-vivo	10854753, 17068095
Rat striatal MSN slice culture (PENK+ GABA)	Engineered in vitro	PENK+ GABAergic Interneuron Activation	(c)	ExperimentalModel	11982623
iPSC MGE/ LHX6+ GABAergic interneurons	Engineered in vitro	PENK+ GABAergic Interneuron Activation	(c)	ExperimentalModel	35269475, 30559206
Human iPSC microglia-like cells	Engineered in vitro	Neuroinflammation; Neurogenic Niche IFN	(a)	ExperimentalModel	41227333, 35803230
Microglia-containing cerebral organoids	Engineered in vitro	Neuroinflammation	(a/b)	ExperimentalModel	39500415
Human BBB-on-chip /	Engineered in vitro	Neuroinflammation	(b)	ExperimentalModel	37818458, 42053998

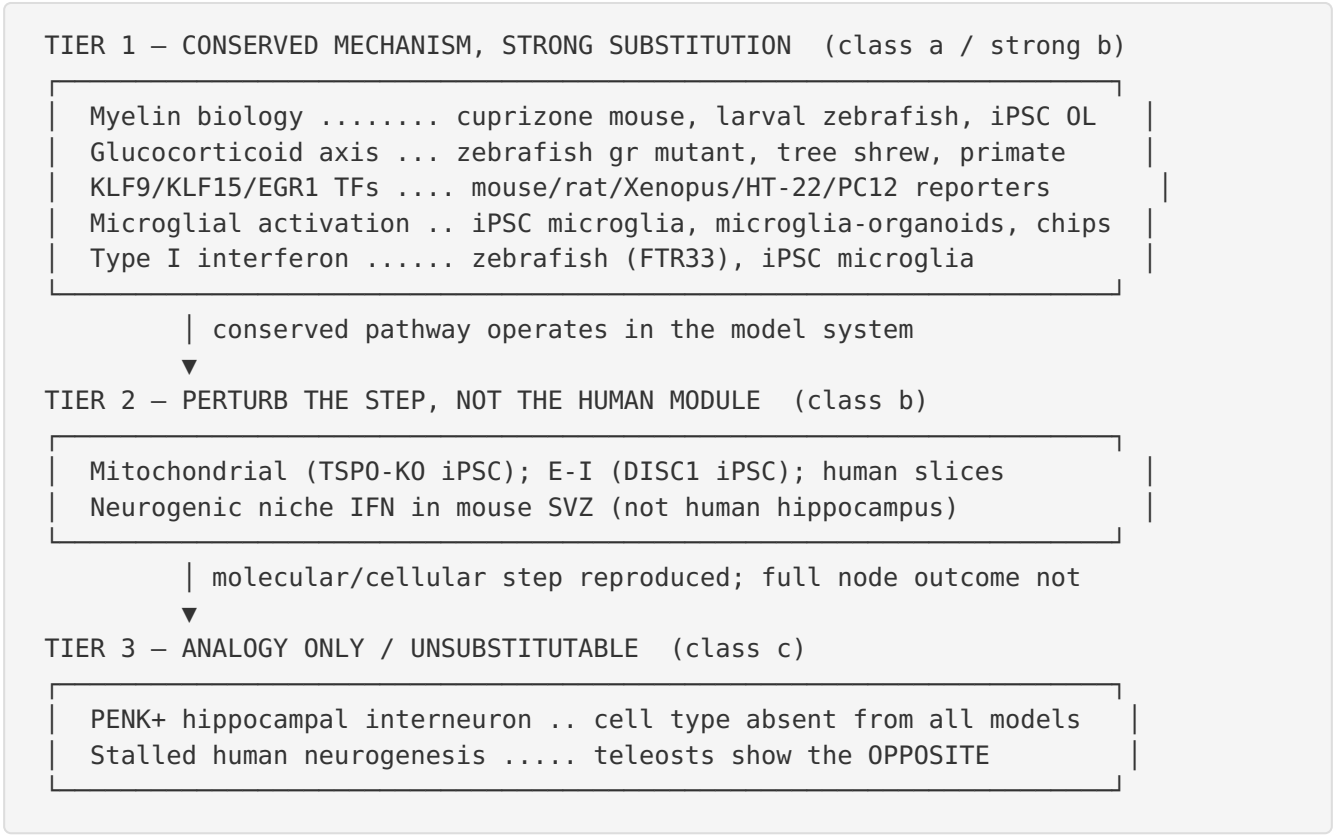
System (species / platform)	Category	Node(s) mapped	Class	Schema block	PMID
neurovascular-unit chip					
Human IFN-α therapy	Controlled exposure (human)	Neuroinflammation	(a)	neither (human induced)	34171555
Human LPS endotoxin challenge	Controlled exposure (human)	Neuroinflammation	(b)	neither (human induced)	34673176
Adult zebrafish chronic unpredictable stress	Controlled exposure	Neuroinflammation; HPA Axis	(a/b)	AnimalModel	40220038
Patient-derived SSRI-resistant MDD iPSC 5-HT neurons	Observational (patient)	Neuroplasticity Deficits; Monoamine Deficiency	(a)	ExperimentalModel	30903001
DISC1-mutant iPSC glutamatergic/ GABAergic co-culture	Genetic	Excitation-Inhibition Imbalance	(b)	ExperimentalModel	38438989
TSPO-KO hiPSC neurons/NPCs/ astrocytes	Genetic	Mitochondrial Dysfunction	(b)	ExperimentalModel	39684592
Adult human organotypic slice culture	Engineered in vitro	Excitation-Inhibition Imbalance; Neuroplasticity	(b/c)	ExperimentalModel	42750245
Fetal human organotypic	Engineered in vitro	Excitation-Inhibition Imbalance	(b)	ExperimentalModel	42214744

System (species / platform)	Category	Node(s) mapped	Class	Schema block	PMID
cortical culture + shRNA					
Tree shrew chronic psychosocial stress	Controlled exposure	HPA Axis Dysregulation	(a)	AnimalModel	15744220
Cynomolgus monkey natural/social-defeat depression	Observational	HPA Axis Dysregulation	(a/c)	AnimalModel	41102858

Empty systems (useful negatives): *Drosophila* and additional *C. elegans* models return nothing for all five focus nodes — they do not myelinate (ruling out the oligodendrocyte node) and lack the relevant GABAergic hippocampal interneuron and adult-hippocampal-neurogenesis biology. Repeated literature searches for Drosophila myelin/glia models and for human hippocampal organoids reproducing the PENK+ population returned no supporting papers; these are reported as search-limited gaps, not proven absences.

Mechanistic Model / Interpretation

The central interpretive problem is the **substitution gap** between evolutionarily conserved biology and human-tissue-defined modules. The candidate landscape sorts into three tiers:



The pattern is coherent: **where the node names a conserved molecule or pathway (myelin, cortisol/ GR, KLF/EGR, type I interferon, microglial cytokines), non-mouse and in-vitro systems substitute well.** Where the node names a **human-tissue-defined cell population or niche state** (the PENK+ hippocampal interneuron; the stalled low-rate human hippocampal neurogenesis), substitution fails or is analogy-only, because the defining feature either does not exist in the candidate (PENK+ hippocampal interneuron) or exists in an opposite form (teleost constitutive neurogenesis).

This is exactly what the brief predicted: three focus nodes were defined from human post-mortem hippocampus by single-nucleus transcriptomics and chromatin accessibility, and the substitution question for them is not taxonomic structure-sharing but whether any system can host a mechanism first observed in human tissue. The answer is **partial** — the *molecular step* can be perturbed in conserved systems, but the *integrated human module* cannot yet be reconstituted.

Evidence Base

PMID	Title (abbrev.)	Role in this report	Study type
22641177	*An affective disorder in zebrafish with mutation of the glucocorticoid receptor*	Supports zebrafish `gr` mutant → HPA node (a)	Controlled genetic manipulation
22962255	*Molecular basis for glucocorticoid induction of Klf9...*	KLF9 direct, conserved GC target	Controlled in-vitro/in-vivo
32032606	*Glucocorticoid receptor required for survival through metamorphosis...*	KLF9 GR-dependence conserved to Xenopus	Controlled genetic KO
30578973	*Crosstalk Between GR and EGR1...BDNF Transcript 4*	Links EGR1 node to neuroplasticity	Controlled manipulation
34544838	*CNS Hypomyelination...Larval Zebrafish*	Zebrafish myelin model (a)	Genetic / imaging
40364981	*Human iPSC...Oligodendrocyte Co-culture...Myelination*	Human in-vitro myelination platform	Engineered in vitro
33972529	*LRIG1 is a gatekeeper to exit from quiescence in adult NSCs*	IFN signature in neurogenic niche	Controlled / observational
29902500	*Neuro-epithelial-like and radial-glial stem/progenitor cells...*	Teleost constitutive neurogenesis = analogy only	Review/ observational
10854753	*Cortical interneurons expressing mu-opioid...*	PENK+ GABA cell type in neocortex	Observational
11982623	*Intrinsic GABA neurons inhibit proenkephalin...neostriatum*	PENK+ GABA MSN outside hippocampus	Controlled ex-vivo
41227333	*Differential Responses of Human iPSC-Derived Microglia...*	Human microglia recapitulate inflammation	Controlled in vitro
34171555	*Influence of gender on cytokine induced depression...*	IFN-α as human induced model	Observational (human)
30903001	*Altered serotonergic circuitry in SSRI-resistant MDD...neurons*	Patient-derived iPSC neuroplasticity phenotype	Observational (patient)

PMID	Title (abbrev.)	Role in this report	Study type
40220038	*Chronic stress...brain monoamines and affective behavior in zebrafish*	Non-mouse neuroinflammation/HPA model	Controlled exposure
38438989	*DISC1 co-cultures reveal aberrant GABAergic phenotypes*	E-I imbalance in vitro	Controlled genetic
39684592	*Human TSPO knockout neuronal model*	Mitochondrial node in vitro	Controlled genetic
42214744	*Human brain slice culture model of STXBP1 encephalopathy*	Perturbable human tissue	Controlled genetic

Correlational vs. controlled: The zebrafish `gr` mutant, Xenopus GR-KO, cuprizone exposure, `mbp:NTR` ablation, DISC1 and TSPO knockouts, and STXBP1 knockdown are **controlled manipulations**. The patient-derived iPSC serotonergic neuron phenotype, the cynomolgus natural-depression model, the IFN-α therapy cohort and the neocortical/striatal PENK+ cell-type descriptions are **observational/correlational** with respect to the MDD module.

Limitations and Knowledge Gaps

1. **The PENK+ GABAergic hippocampal interneuron node is unsubstitutable.** No model recapitulates it. The homologous cell type exists in neocortex and striatum but the specific human hippocampal population has not been identified in any model, including human iPSC interneuron systems. This is the single most important negative result.
2. **Human-tissue-defined modules resist in-vitro reconstitution.** Conserved systems perturb the *molecular step* (KLF/EGR induction, type I interferon, microglial cytokines) but none reconstitutes the integrated human hippocampal multi-cell module or the chromatin-accessibility landscape from which nodes 10–12 were defined.
3. **Behavioural-proxy vs. molecular-signature conflation.** Several otherwise strong models (cuprizone mouse, chronic-stress zebrafish, tree shrew, cynomolgus) read out depression-like *behaviour*, which is not evidence about a transcriptional module. This distinction was enforced throughout but constrains how far these models can be claimed to "support" the human-defined focus nodes.

4. **Adult hippocampal neurogenesis is contested in humans and divergent across candidates.** Mouse NSC work is in the SVZ, not hippocampus; teleosts show the opposite (constitutive) phenotype. No model cleanly reproduces "stalled human hippocampal neurogenesis."
5. **Search-limited gaps, not proven absences.** Searches for Drosophila myelin/glia models and for human hippocampal organoids reproducing the PENK+ population returned no papers. These are reported as gaps in the available literature, not as demonstrated non-existence.
6. **Node-scale ambiguity.** Where node scales were stated as "not stated," I inferred the level from each node's description (E-I Imbalance → cellular/circuit; TF Reprogramming → molecular; Oligodendrocyte → cellular/tissue). Different scale assignments could change which models count as recapitulating.

Proposed Follow-up Experiments / Actions

1. **Target the PENK+ hippocampal interneuron directly.** Perform single-nucleus RNA-seq on human hippocampal organoids/assembloids and on long-term human organotypic hippocampal slice cultures to test whether a PENK+ GABAergic interneuron population arises; this is the only path to moving node 12 from class (c) toward (b) or (a).
2. **Build a human iPSC "KLF9/KLF15/EGR1 glucocorticoid-reporter" line.** Combine GRE/KSM reporters with patient-derived MDD lines and glucocorticoid challenge to reconstitute the TF-reprogramming step in a human hippocampal-like context, bridging the conserved rodent/Xenopus evidence to the human module.
3. **Interferon challenge of microglia-containing hippocampal organoids.** Combine the microglia-in-organoid platform with type I interferon stimulation and neurogenic-niche readouts to test node 10 in a human neurogenic context rather than mouse SVZ.
4. **Couple a demyelination manipulation to transcriptional (not behavioural) readout.** Apply cuprizone/lysolecithin or `mbp:NTR` ablation and profile oligodendrocyte-lineage transcriptomes against the human hippocampal signature, decoupling the myelin lesion from the behavioural proxy.

5. **Curate the clear wins now.** The zebrafish `gr` mutant (AnimalModel → HPA), cuprizone mouse and larval zebrafish (AnimalModel → Oligodendrocyte), human iPSC microglia and microglia-organoids (ExperimentalModel → Neuroinflammation), and iPSC OL co-cultures (ExperimentalModel → Oligodendrocyte) are defensible additions that can be entered into the schema immediately, each with its stated limitation.
6. **Record the negatives.** Enter the Drosophila/*C. elegans* myelin exclusion and the PENK+ hippocampal-interneuron substitution failure as documented negative results, so future curation does not re-propose them.
